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Numerical Study on the Influence of CO₂ Dilution on Soot Emission in Turbulent Diffusion Flames Impinging on a Cold Surface

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ABSTRACT

A flame impinging on a cold surface, frequently utilized in engineering applications, produces a greater amount of soot, which affects health and environmental impacts. The CO₂-diluted turbulent nonpremixed flame impinging on a cold surface is computationally investigated because it is one of the common techniques for reducing emissions. The amount of soot is compared between the nondiluted flame and the flame with 25%, 37.5%, and 50% CO₂ dilution by volume. The study finds that a higher dilution ratio results in less soot being emitted for the same amount of fuel used. To study the chemical effect of CO₂, nonreactive fictitious CO₂ (FCO₂) is simulated. The results reveal that while the reduction in flame temperature under higher dilution ratios heavily suppresses soot formation processes thermally, the chemical action of CO₂ also provides an important mechanism, accounting for an additional 14.3%, 19.5%, and 58.4% of emission reduction at 25%, 37.5%, and 50% dilution, respectively. As the dilution ratio increases, all soot production mechanisms decrease quantitatively. Furthermore, a model-based mechanistic hypothesis for the chemical influence of CO₂ addition is proposed, suggesting that it reacts with the H radical. The reduction in H radicals reduces the amount of polycyclic aromatic hydrocarbons (PAHs), which are precursors of soot particles, resulting in lower emissions. Finally, the power that is generated is considered while evaluating soot reduction performance. The CO₂ dilution is found to be more effective at greater dilution ratios while maintaining the same heat transfer rate.

1 | Introduction

Researchers have become increasingly interested in soot emissions due to their harmful effects. Particulate matter has been found as a cause of disease, including cardiovascular mortality and lung cancer [1]. This carbonaceous particle is mostly produced during the burning of hydrocarbon fuels [2], especially with nonpremixed flames, which are known to emit a lot of soot. Particles smaller than 2.5 μm (PM_{2.5}) are more hazardous as they can reach the respiratory system [3].

Several human activities rely on hydrocarbon combustion, including cooking. According to a World Health Organization (WHO) survey in 2025, a substantial portion of the global population uses biomass cookstoves [4–6]. This process involves incomplete combustion, generating a significant amount of soot [7], which is implicated as a cause of millions of deaths annually [6]. Consequently, developing advanced high-fidelity numerical models to track the production, growth, and mitigation of soot aggregates remains a major focus of modern combustion science [8]. Previous experimental and numerical studies reveal that large amount of soot is emitted as the flame contacts

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the cooking ware [9, 10]. The impingement of the flame on a cold surface is not limited to the cooking process, since there are many industrial applications utilizing such a setup, including the heating and melting of glass and metal [11]. Flame impingement heating provides a large increase in heat transfer rate [12]; however, it also produces a lot of soot. To ameliorate the situation, stricter rules have been adopted to limit emissions [2]. This configuration was analyzed numerically, revealing that the scalar dissipation rate is the primary parameter affecting the emission behavior [13].

Several experiments have been undertaken to investigate the effects of CO₂ dilution on soot behavior. In 1968, McIntock et al. report that introducing CO₂ into the oxidant stream can successfully cut emissions [14]. Schug et al. [15] state in their work that the ability to minimize soot from CO₂ dilution is exclusively attributable to the thermal effect. However, Du et al. [16] propose that CO₂ chemically lowers emissions when added to both the fuel and oxidizer streams. According to Zhang et al. [17], CO₂ chemically raises OH concentration, which attenuates the nucleation process and reduces soot. According to Angrill et al. [18] CO₂ dilution impacts soot formation but not the oxidation process. Guo and Smallwood [19] quantitatively investigate CO₂ addition in laminar co-flow nonpremixed flames and find that dilution lowers soot production chemically. The study suggests that the PAH formation and soot surface growth process are attenuated as the concentration of H radical is reduced. Wang and Chung [20] experimentally and computationally examine the chemical influence of CO₂ dilution in ethylene counterflow diffusion flame by comparing it to N₂ addition. Liu et al. [21] find that CO₂ dilution leads to lower flame temperature, which reduces soot. However, due to the difference in heat capacity between both species, the chemical effect of CO₂ is hardly quantified [22]. Ying and Liu [23] report that CO₂ dilution suppresses soot formation through the nucleation process in the inverse diffusion flame, while Duan et al. [24] find that CO₂ addition promotes soot oxidation reactivity. Although the literature broadly agrees that CO₂ dilution effectively reduces soot emissions under many conditions, the relative dominance and interplay among individual mechanisms—including physical dilution, thermal cooling, radiation, H/OH radical chemistry, PAH suppression, and oxidation pathways—are highly contingent upon the specific combustion configuration [25]. Mapping out how these competing roles shift under different flow and thermal boundary conditions is essential to create a cleaner combustion process.

To reconcile high heat transfer demands with stringent environmental regulations, CO₂ dilution has emerged as a key technique in advanced combustion systems. The introduction of CO₂ into the fuel or oxidizer streams is highly relevant to several practical engineering configurations: exhaust gas recirculation (EGR), modern internal combustion engines, and industrial burners recirculate cooled, CO₂-rich exhaust gases back into the combustion chamber to suppress flame temperatures and lower NO_x emissions, which significantly affects soot inception kinetics. Advanced Cookstoves: Modern low-emission biomass cookstoves intentionally use secondary internal gas recirculation zones to carry inert combustion products back into the reaction sheet to limit soot formation. While recent work has systematically decoupled the thermal and chemical effects of CO₂ dilution in basic laminar or partially premixed free flames [26–28], the

multiphysical behavior of diluted turbulent streams under severe surface-impingement quenching is not yet fully understood.

While many applications relate to it, particularly those involving direct impinging heat transmission, research on the emission behaviors in diffusion impinging flames is quite limited in comparison to those of free flames [13]. Previous research indicates that the contact of a flame with a cold surface alters both the flame structure and the soot emission characteristics compared to its free flame counterpart [9, 13]. Specifically, a primary short-coming of previous studies on CO₂ soot suppression is their focus on unconfined free flames, which leaves unresolved how near-wall thermal quenching and high scalar dissipation modify these chemical pathways. Therefore, this study aims to numerically investigate the emission characteristics in such a configuration. As well as the influences reported on the free flame, this work hypothesizes that CO₂ dilution could suppress the soot formation of the nonpremixed impinging flame through both nonchemical and chemical effects. Although experimentation offers a direct method for data collection, it is subject to certain limitations. This work employs a numerical technique because it enables the isolation of the chemical effects of CO₂, which is impractical in experimental settings. The current computational study focuses on the impact of CO₂ dilution on soot emission from an ethylene/air turbulent diffusion flame impinging on a cold surface. The soot behavior is simulated as the dilution ratio changes. Fictitious CO₂ (FCO₂) is employed for the first time to decouple the chemical and thermal effects resulting from CO₂ dilution in the fuel stream, specifically in a complex turbulent nonpremixed impinging flame. This marks a new point of our work by extending species-decoupling techniques from laminar flames into wall-bounded turbulent environments. The effect of dilution on soot behavior is investigated both qualitatively and quantitatively. The sensitivity of soot formation to the dilution ratio is illustrated through an evaluation of the normalized volume-integrated source terms. Moreover, this study explores the applicability of this mechanism within a complex turbulent impinging flame configuration, supporting existing literature insights under localized cooling constraints. Finally, this study utilizes an application-relevant engineering metric by normalizing the emission factor with the heat transfer rate, thereby offering a practical approach to optimizing impinging flame dilution where a balance between emissions and thermal efficiency is required.

2 | Soot Dynamics Model

In this work, the experimental flame from the study of Pundle [9] is computationally investigated. The experiment investigated ethylene, a commonly utilized fuel for sooting flames impinging on a cool surface. To replicate the experiment, a two-dimensional axisymmetric turbulent nonpremixed flame is simulated using ANSYS Fluent on structured grids. Grid independence has been confirmed. A jet of ethylene at 47 m/s with the temperature of 300 K was injected into the quiescent air via a 2.18 mm diameter nozzle, resulting in a Reynolds number of 11,300. In the experiment, a constant 373 K surface with a diameter of 240 mm and a height of 140 mm is used to reflect a cooking pot filled with boiling water. According to earlier experimental [9] and computational investigations [13], the simulation schematic diagram is depicted in Figure 1, where the fuel flows in the upward

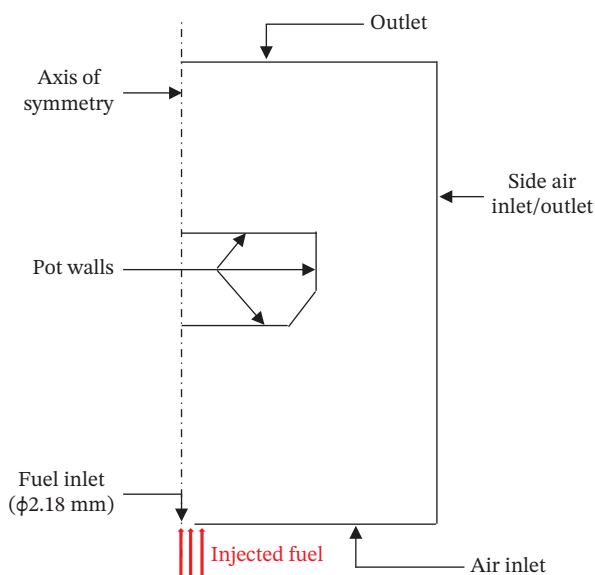


FIGURE 1 | Computational domain and boundary conditions [9].

direction through the nozzle indicated as fuel inlet, while the ambient air flows (in and out) through the boundaries indicated as air inlet, side air inlet/outlet, and the exhausted gas leaves the system at the outlet.

This work uses the commonly utilized sectional approach to simulate the dynamics and evolution of the soot particle size distribution function (PSDF) [29–31]. In general, the sectional technique discretizes the PSDF by dividing it into several size sections before solving the well-known particle population balance equation (PBE) for each section, which requires extensive resources. To address this issue, the current study used the fixed-pivot sectional approach developed by Kumar and Ramkrishna [32] in 1996, which has been used in numerous studies [10, 13, 31, 33–35]. The simulation includes 30 size sections with a geometric volume spacing factor (f_v) of 2, which encompass the diameter of soot particles ranging from 0.89 to 726 nm. The fixed-pivot sectional method is incorporated into ANSYS Fluent by creating 30 user-defined scalars (UDS) where each UDS explicitly tracks the particle number density (particles per unit volume, m^{-3}) within its respective discrete particle volume section. More details can be found in [10]. To solve mass, momentum, and energy conservation equations, the Reynolds Averaged Navier–Stokes Equation (RANS) is utilized. The SST $k-\omega$ model is used in this study because it incorporates a blending function that transitions between the $k-\omega$ model in the near-wall region and the $k-\epsilon$ model in the free-stream zone, both of which are essential for this work. This study employs the established chemical mechanism known as the *FORCE*, developed by [36], to ensure thermodynamic consistency and prevent inaccuracies in soot behavior prediction. The *FORCE* includes the PAH pathways essential for the soot nucleation process. Such a mechanism includes 174 species and 1896 reactions. Moreover, to address the turbulent chemistry interaction, this work employs the established flamelet model, which considers the turbulent flame as an aggregate of one-dimensional, thin, and laminar flame structures. Such an approach calculates species concentrations that are precalculated as a function of scalar dissipation and mixture fraction and tabulated as a table [37].

In general, four key processes influence the behavior of soot particles: nucleation, surface development, coagulation, and oxidation [38]. All processes are used to calculate the source/sink term in the scalar transport equation for the mentioned UDS. In the current study, nucleation is based on dimerization between 2 pyrene molecules [31], while surface growth employs the conventional H-abstraction- C_2H_2 -addition mechanism (HACA) [38] along with the concept of active surface sites. The coagulation process is modeled using the approach developed for the fix-pivoted sectional method described in [32], which is based on the Smoluchowski equation. Soot oxidation by OH is simulated using the work of Neoh et al. [39], which is based on the collision efficiency, whereas oxidation by O_2 is based on the prior work of Charoensin-O-larn et al. [10] which also depends on the active surface site. Because the presence of soot particles alters the temperature field via the radiation process, the discrete ordinates method (DO) is used, coupled with the absorption coefficient described in [40]. Radiation absorption and emission by the gas-phase species (including CO_2 and FCO_2) were neglected, as radiative heat transfer in highly sooting ethylene flames is overwhelmingly dominated by soot particles. The model utilized in this work has been validated against experimental flames reported in the literature, specifically referenced in [10] for further details. Moreover, the model employed in this study has been demonstrated close agreement with the experimental results of the soot emission factor (SEF) as shown in Figure 2, where the distance between the fuel nozzle and the surface ranges between 150 and 300 mm. However, this work focuses on the 245 mm case, which yields the highest SEF. The reader can refer to [13] for more information on the modeling details. While the model has been validated against measured SEFs, it is important to note that experimental data for local flame shape, temperature distribution, and soot volume fraction in impinging flame configurations are not available for direct comparison with this particular impinging configuration. This constrains the validation of spatially resolved predictions and underscores the necessity for forthcoming experimental research to acquire such data. Consequently, the spatial fields, radical concentrations, and chemical source term trends discussed herein are presented as model-based mechanistic hypotheses to guide future experimental verification rather than absolute validated predictions.

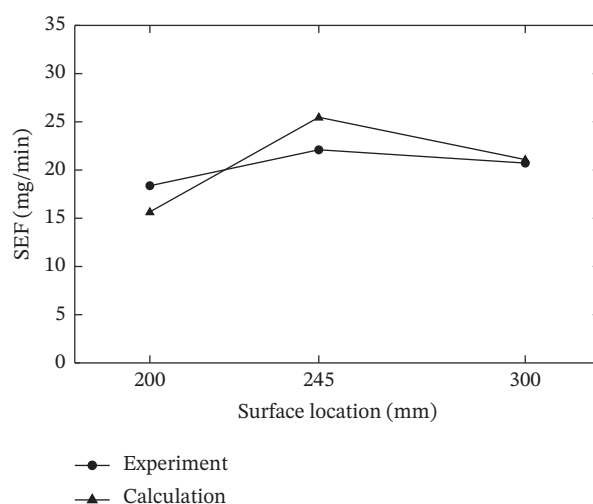


FIGURE 2 | The plot of the soot emission factor (mg/min) with variation of distance between the fuel nozzle and the surface [13].

TABLE 1 | Relative errors of the soot emission factor (mg/min) obtained from different meshes.

No.	Number of elements	Soot emission factor (mg/min)	Relative error (%)
1	86,274	24.118	—
2	103,417	25.172	4.37
3	124,352	25.408	0.94

According to the previous section, to evaluate the effect of CO₂ dilution, a fuel stream containing ethylene and CO₂ is simulated with dilution ratios of 25%, 37.5%, and 50% by volume. As previously stated, there is no consensus on how dilution with CO₂ chemically suppresses soot formation; so, due to the benefit of computational modeling, nonreactive fictitious CO₂ (FCO₂) is introduced into the simulation. To elaborate, the FCO₂ has been added to the thermodynamics database with identical properties to CO₂. However, in contrast with CO₂, there is no reaction in the chemical mechanism (the FORCE) that contains FCO₂ since the species does not react. This method was recently developed and has been utilized in a few investigations with various flame configurations, including the work of Mahmoud et al. [41]. The chemical influence can be determined by comparing the results of CO₂ and FCO₂ additions, as FCO₂ can only reduce soot production through thermal and dilution effects. As a result, the chemical effect is responsible for the additional soot suppression.

3 | Results and Discussion

Following the previous works [9, 10], the SEF (mg/min) is defined to reflect the overall quantity of soot emitted by the flame. Such a parameter can be determined from a surface integration at a cross-sectional area at the outlet using the expression

$$SEF = \int f_v \rho_{\text{soot}} v_{\text{axial}} dA, \quad (1)$$

where f_v is the soot volume fraction, ρ_{soot} is the soot particle density (1.8×10^9 mg/m³), and v_{axial} (m/min) is the local axial flow velocity. To obtain grid independence, three meshes were utilized for the undiluted case, with element counts of 86,274, 103,417, and 124,352. Table 1 portrays the SEF along with its associated relative errors for the three meshes. Due to the minimal relative errors between mesh number 2 and mesh number 3, the mesh consisting of 103,417 elements is employed for all simulations in this study.

Figure 3 shows a variation in the SEFs on a logarithmic scale to clearly evaluate the difference at different CO₂ dilution ratios.

Figure 3 shows that, as anticipated, the SEF decreases with increasing dilution ratios for both CO₂ and FCO₂ dilution. However, the reduction in quantity could be attributed to a decrease in fuel mass, which has a direct impact on the amount of soot formed. To isolate the influence of the dilution ratio, the normalized soot emission factor, NSEF, is introduced, which can be derived using the following expression:

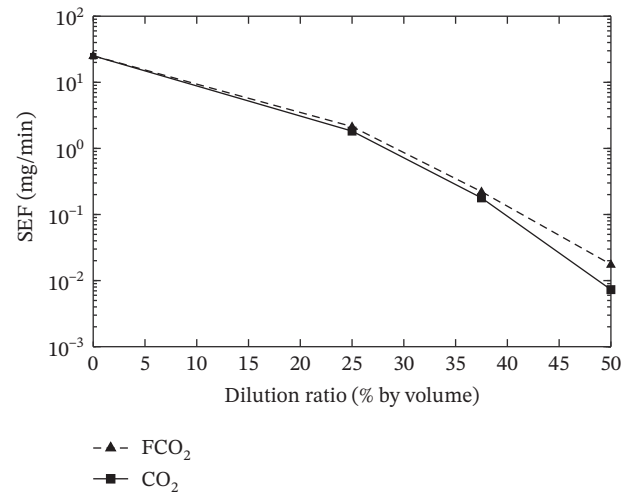


FIGURE 3 | The plot of the soot emission factor (mg/min) at different dilution ratios.

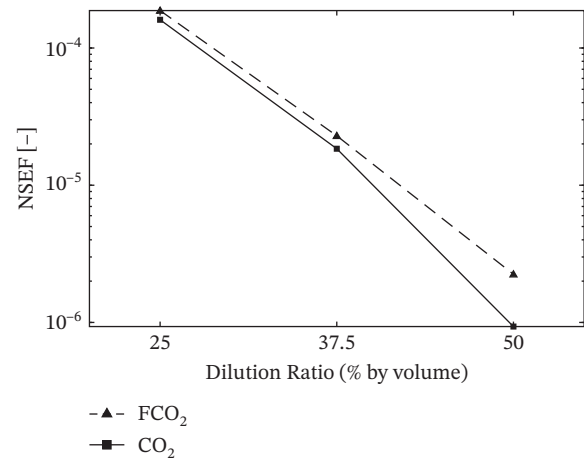


FIGURE 4 | The plot of the normalized soot emission factor at different dilution ratios.

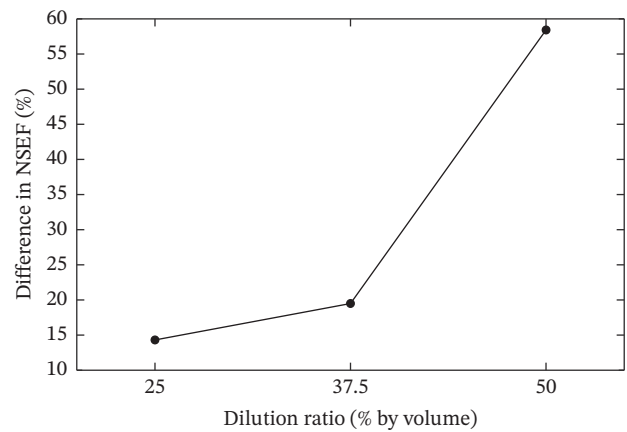


FIGURE 5 | The difference in the normalized soot emission factor between FCO₂ and CO₂ at different dilution ratios.

$$NSEF = \frac{\text{soot emission factor}}{\dot{m}_{C,\text{fuel}}}, \quad (2)$$

where $\dot{m}_{C, \text{fuel}}$ is the mass flow rate of reacting carbon atoms in the fuel stream which is contributed by C_2H_4 . Using this quantity, the effect of dilution can be assessed independently of the fuel mass effect. Figure 4 depicts the fluctuation of the normalized soot emission factor (NSEF) on a logarithmic scale at various dilution ratios. The nondiluted flame data is not given here to highlight the difference in dilution ratios.

Figure 4 shows that the influence of carbon atom mass is eliminated; the FCO_2 - and CO_2 -diluted flames follow the same trend, with the NSEF decreasing as the dilution ratio increases. This

means that with higher dilution ratios, less soot is created for the same amount of fuel combusted. Furthermore, the CO_2 -diluted flame produces less soot than the FCO_2 -diluted one. This implies that introducing CO_2 has a chemical effect on soot production. Figure 5 depicts the percentage difference in the NSEF between FCO_2 and CO_2 , which helps to better understand the chemical effect. Such a percentage difference is calculated based on the expression:

$$\text{Difference (\%)} = \frac{(\text{Normalized soot emission factor})_{FCO_2} - (\text{Normalized soot emission factor})_{CO_2}}{(\text{Normalized soot emission factor})_{FCO_2}} \times 100. \quad (3)$$

Figure 5 shows that the chemical effect of CO_2 dilution reduces the amount of soot emission compared to FCO_2 dilution. It also demonstrates that the effect of the chemical reaction rises nonlinearly with the CO_2 addition ratio. According to the plot, at 50% dilution, the chemical impact can suppress 58.4% of the emission, while the differences at 25% and 37.5% dilution are 14.3% and

19.5%, respectively. Furthermore, there is a notable increase in the difference from 37.5% dilution to 50% dilution. This occurs due to elevated dilution ratios; soot quantity decreases, leading to an increased sensitivity to the concentration of precursor species, which reflects the chemical effect of CO_2 . It is important to note, however, that at the highest dilution ratio (50%), the SEFs are

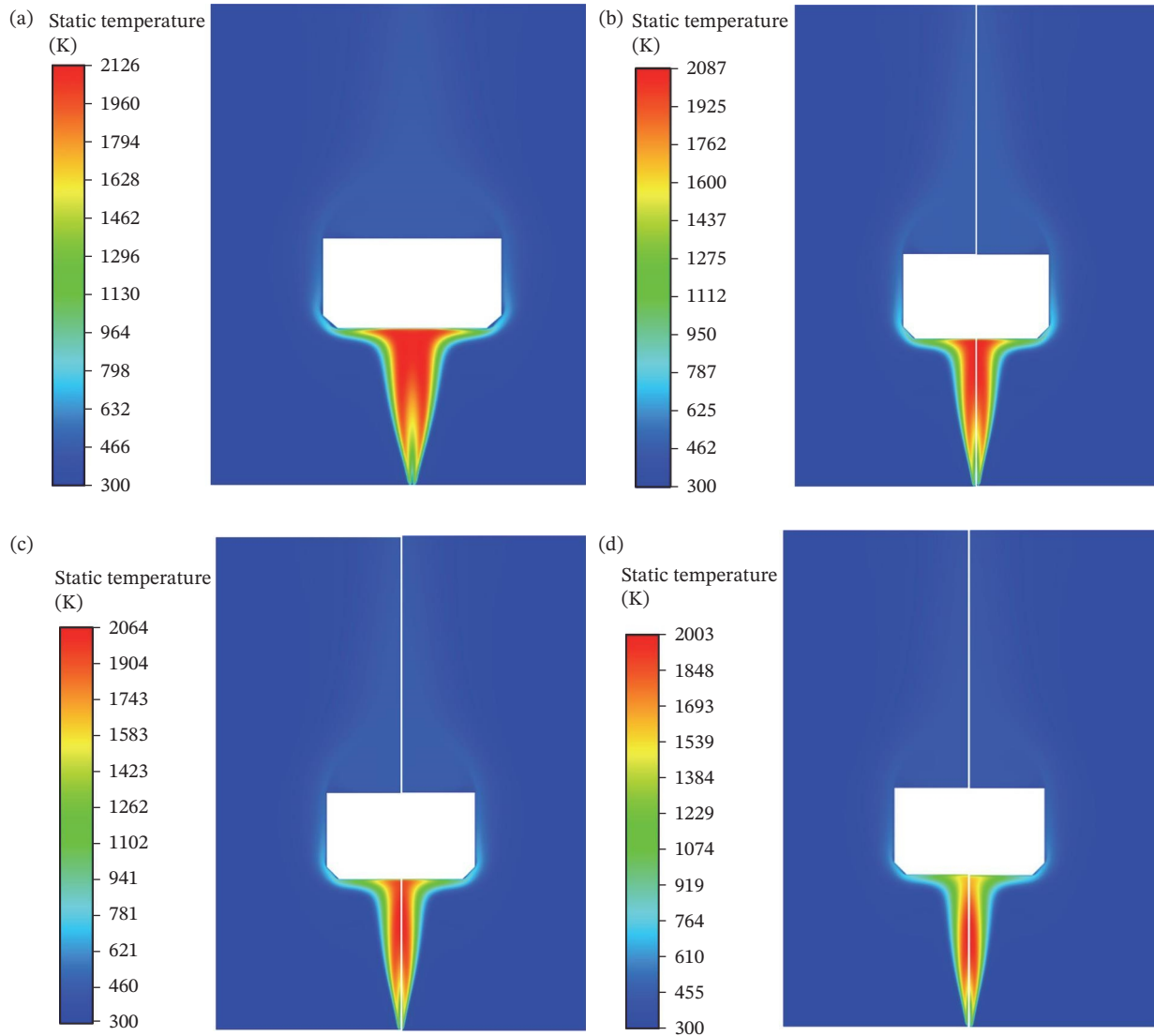


FIGURE 6 | Temperature contour plot of CO_2 - (left) and FCO_2 - (right) diluted cases at the dilution ratio of (a) 0%, (b) 25%, (c) 37.5%, and (d) 50%.

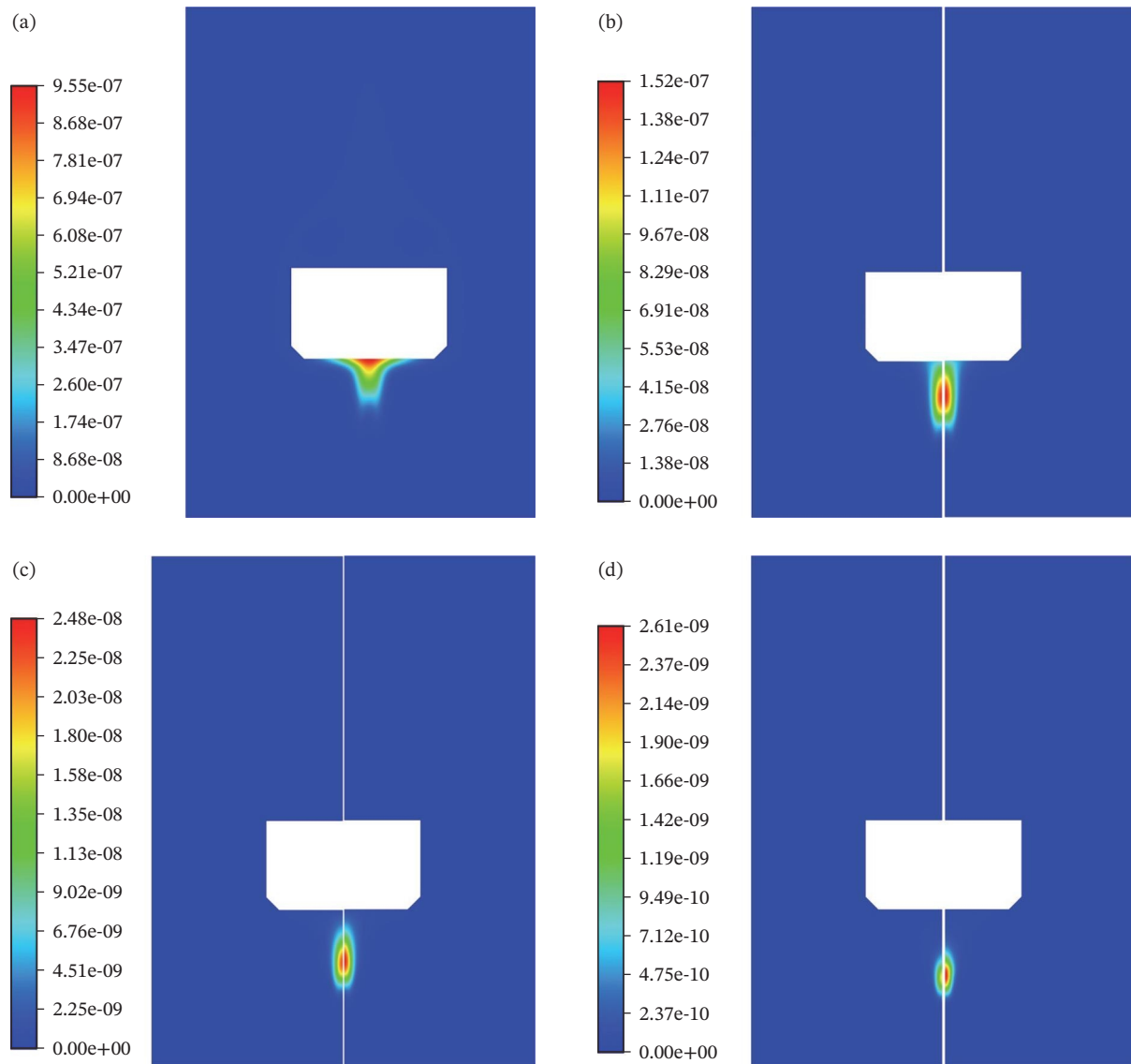


FIGURE 7 | Soot volume fraction contour plot of CO_2 - (left) and FCO_2 - (right) diluted cases at the dilution ratio of (a) 0%, (b) 25%, (c) 37.5%, and (d) 50%.

suppressed by several orders of magnitude compared to the baseline undiluted flame. Consequently, while the relative chemical suppression effect of CO_2 appears mathematically dominant at 58.4%, this value represents a variation between two very low absolute quantities. At these near-zero soot thresholds, the numerical results become highly sensitive to model uncertainties. Therefore, while the trend firmly establishes that the chemical effect grows in relative importance as dilution increases, cautious interpretation is required regarding the exact numerical percentage at extreme dilution boundaries due to these coupled numerical and kinetic sensitivities.

To illustrate the impact of the dilution ratio, Figures 6 and 7 display the contour plots of the temperature field and soot volume fraction, respectively, for both CO_2 - and FCO_2 -diluted cases.

As expected, the undiluted flame yields the maximum localized heat transfer rate due to its elevated flame temperature (Figure 6) and produces the highest SEF (Figure 7). Moreover, Figure 6 illustrates that temperature diminishes with an increase in dilution ratio for both CO_2 and FCO_2 cases due to a smaller amount of fuel being combusted. The temperature fields are comparable

for the CO_2 - and FCO_2 -diluted cases at matching dilution ratios. Another critical parameter is soot volume fraction, as it closely correlates with the quantity of soot emitted. Figure 7 depicts that an increase in the dilution ratio reduces the soot volume fraction for both diluting species, as expected. Nonetheless, a notable behavior occurs as the peak of the soot volume fraction shifts

TABLE 2 | Quantitative comparison of peak flame temperatures (T_{max}) for the evaluated configurations.

Dilution ratio (%)	Diluting species	T_{max} (K)	Difference (%)
25	CO_2	2085	0.10
	FCO_2	2087	
37.5	CO_2	2054	0.48
	FCO_2	2064	
50	CO_2	1987	0.75
	FCO_2	2002	

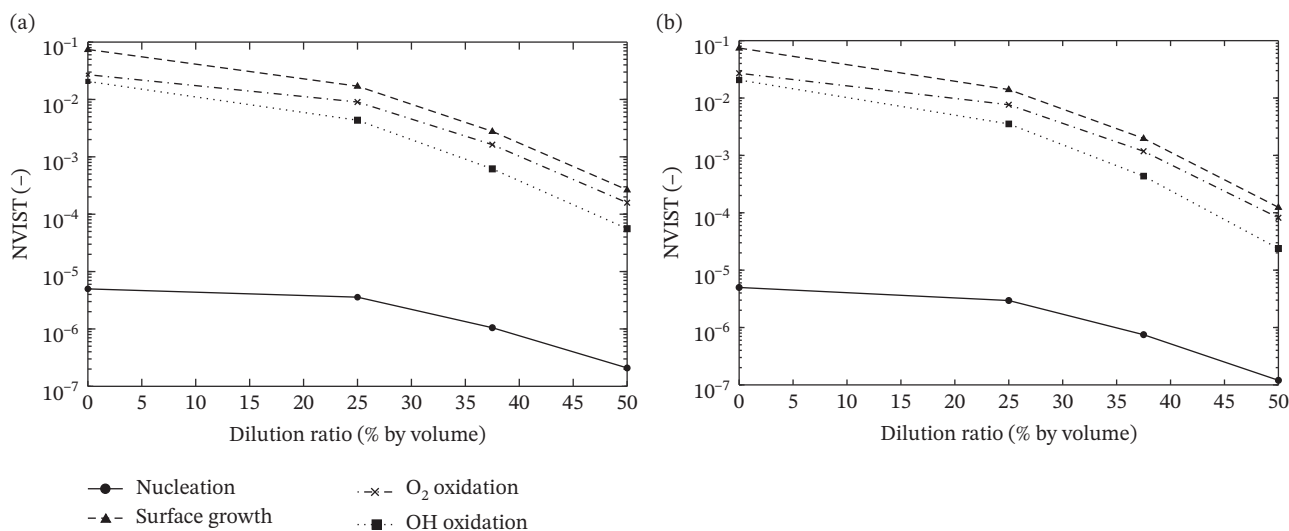


FIGURE 8 | The normalized volume integration of source terms for (a) FCO₂ and (b) CO₂ dilution.

TABLE 3 | Relative percentage changes in normalized volume-integrated source terms.

Diluting species	Increase of dilution ratios (%)	% Change			
		Nucleation	Surface growth	O ₂ oxidation	OH oxidation
FCO ₂	0 to 25	−28.24	−77.06	−66.61	−78.91
	25 to 37.5	−70.66	−83.72	−81.94	−85.81
	37.5 to 50	−80.08	−90.43	−90.27	−90.93
CO ₂	0 to 25	−40.79	−81.09	−71.74	−82.84
	25 to 37.5	−74.63	−85.94	−84.54	−87.69
	37.5 to 50	−83.95	−93.70	−93.07	−94.53

toward the fuel nozzle with an increase in the dilution ratio. This is because the flame is shorter for the higher dilution ratio, as there is less fuel to combust. Moreover, at the same dilution ratio, the soot volume fraction is higher in the case of FCO₂ dilution, which is consistent with Figure 3. To rigorously isolate the thermal, dilution, and chemical pathways, the global flame structures for both the reactive CO₂ and FCO₂ cases must be quantitatively evaluated. Table 2 provides a comparative summary of the peak flame temperature ($T_{\max}$) across the investigated dilution range.

As shown in Table 2, increasing the dilution ratio under a constant volumetric flow rate results in a monotonic decrease in peak flame temperature for both cases, illustrating the progressive thermal and physical quenching effect of the diluent stream. Crucially, when comparing the CO₂ case to its nonreactive FCO₂ counterpart at any fixed dilution level, the peak temperatures remain remarkably close, varying by a maximum of only 15 K (less than 0.8% variance) at the highest dilution. This precise quantification provides rigid justification that the physical dilution and thermal field configurations are successfully isolated by the FCO₂ framework, confirming that the remaining variations in soot kinetics are driven strictly by the chemical effects of CO₂.

As stated earlier, understanding the mechanisms underlying this behavior is as crucial as understanding the consequences. Because the amount of soot emission is determined by the

cumulative effect of soot generation processes over the simulation domain, each process volume-integrated source term is calculated. To reduce the influence of carbon atom mass in the fuel, the dimensionless normalized volume-integrated source terms, NVIST of FCO₂- and CO₂-diluted flames are determined using the equation as expressed:

$$\text{NVIST} = \frac{\text{Volume integration of source term}}{\dot{m}_{\text{C,fuel}}}, \quad (4)$$

where $\dot{m}_{\text{C,fuel}}$ is the mass of reacting carbon in the fuel stream. Figure 8 depicts the logarithmic-scaled normalized volume integration of source terms for both FCO₂- and CO₂-diluted flames with varying dilution ratios.

Both plots of Figure 8 show that all source terms decline as the dilution ratio increases for both the flame with FCO₂ and CO₂ dilution. To easily evaluate the impact of the dilution by both species on source terms, Table 3 presents the relative percentage changes in normalized volume-integrated source terms.

Table 3 indicates that a greater relative percentage change in normalized volume-integrated source terms occurs at higher dilution ratios for both diluting species. Moreover, CO₂ surpasses FCO₂ in lowering the soot source parameters, corroborating earlier results. Nucleation and surface growth are mass-addition

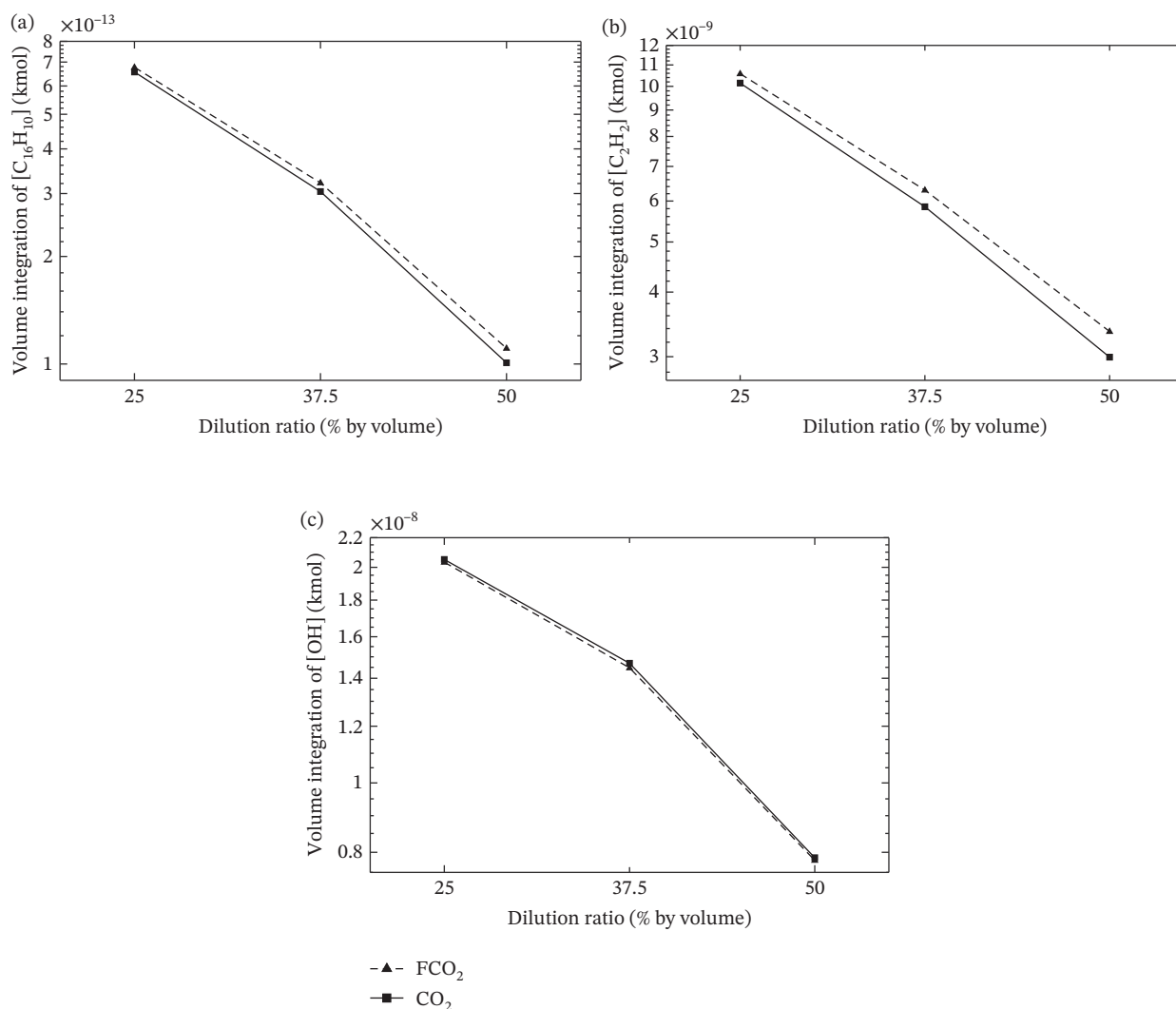
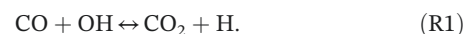


FIGURE 9 | The volume integration of (a) pyrene, (b) C_2H_2 , and (c) OH concentrations.

processes, but both oxidation processes remove soot mass. As previously stated, the inception process is dependent on pyrene concentration, while the surface growth process requires C_2H_2 . The concentration of OH determines the rate of OH oxidation. According to Table 3, the oxidation rate of OH escalates at increasing rates as the dilution ratio increases for both diluents. Nonetheless, the CO_2 -diluted flames consistently demonstrate a higher relative OH oxidation rate compared to their FCO_2 counterparts. This is consistent with the model-based hypothesis that CO_2 chemically affects the soot production and oxidation process, as the concentration of OH is elevated in CO_2 -diluted flames at the same dilution ratio. Apart from species concentration, a decrease in soot emissions correlates with a reduction in particle surface area, which impacts soot formation processes. Because the oxidizer stream has a high concentration of O_2 , there should be negligible O_2 concentration between the flame with FCO_2 and CO_2 dilution. Hence, it is logical to investigate the behavior of pyrene, C_2H_2 , and OH. Figure 9 depicts the change in volume integration of pyrene, C_2H_2 , and OH concentrations on a logarithmic scale over the simulated domain. The undiluted flame case is not shown here to demonstrate the difference in the diluted situation.

Figure 9 shows that CO_2 chemically suppresses pyrene and C_2H_2 , which are essential for nucleation and surface growth processes, respectively. In contrast, CO_2 dilution causes a slightly larger volume integration of OH concentration. As a result, the CO_2 -diluted flame contains less amount of soot. Moreover, it is worth noting that a larger amount of OH eventually leads to a higher soot oxidation rate under identical conditions. It is also interesting to determine how CO_2 dilution reduces soot. Since soot particles are created from polycyclic aromatic hydrocarbons (PAHs), which are dependent on the amount of H radical, it is logical to assess the reaction that impacts the species concentration. This work considers the reaction suggested in the literature [19]:



To analyze the reaction, Figure 10 displays all volume-integrated species except OH on a logarithmic scale, which is depicted in Figure 9.

As shown in Figures 9 and 10, the CO_2 -diluted flame contains more OH, CO, and CO_2 and less H radical at all dilution ratios. This is because, according to the reaction R1, adding more CO_2 creates more CO and OH while decreasing the concentration of

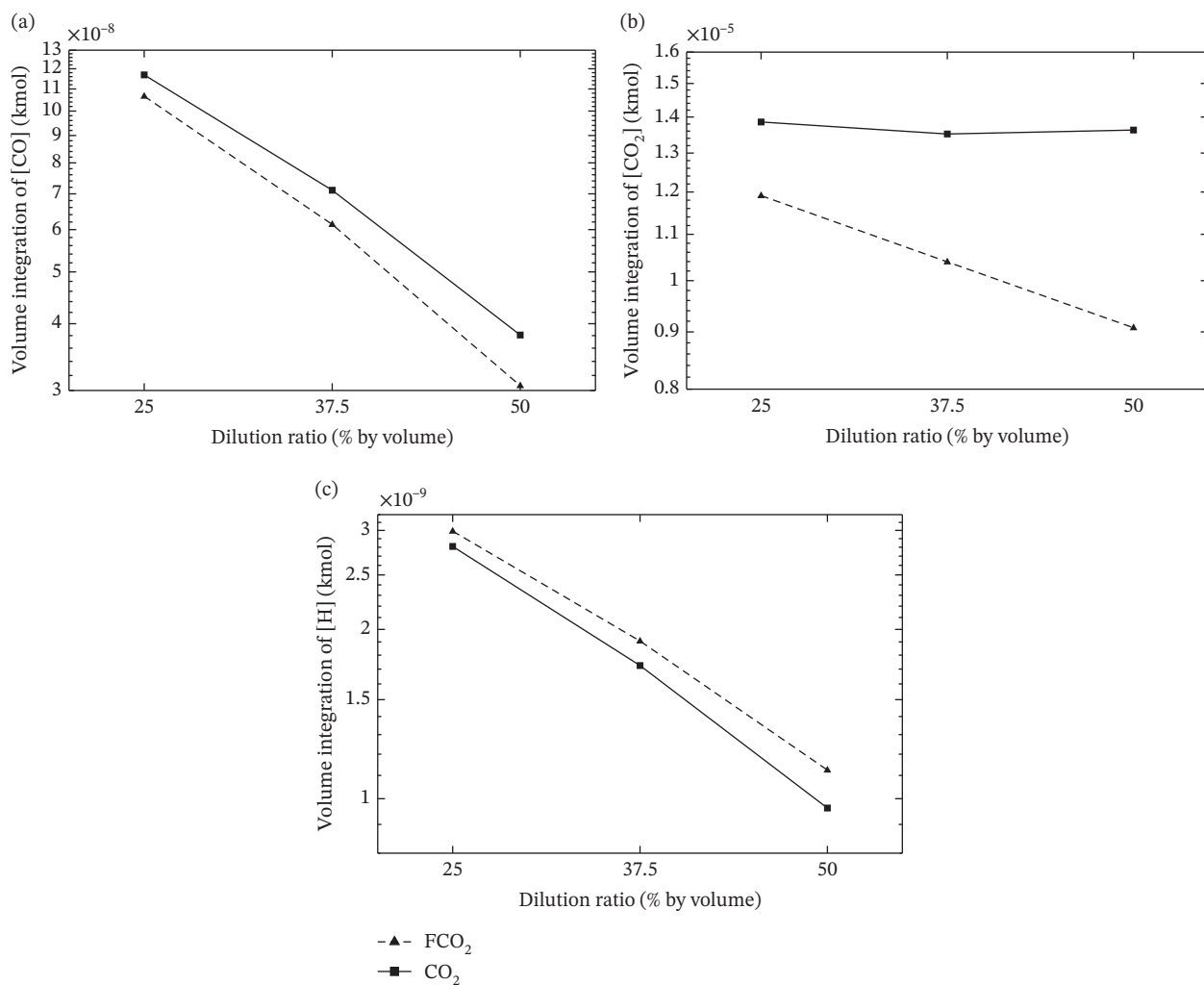


FIGURE 10 | The volume integration of (a) CO, (b) CO₂, and (c) H radical concentrations.

TABLE 4 | The emission factor normalized by the heat transfer rate.

Dilution ratio (%)	Emission factor (mg/min)	Heat transfer rate (kW)	Emission factor/heat transfer rate (mg/kJ)
25 (CO ₂)	1.8270	3.0664	0.00993
25 (FCO ₂)	2.1398	3.0585	0.0117
37.5 (CO ₂)	0.1783	2.1461	0.00138
37.5 (FCO ₂)	0.2216	2.1464	0.00172
50 (CO ₂)	0.0073	1.5476	0.0000786
50 (FCO ₂)	0.0176	1.5482	0.000189

H radical as the reaction shifts to the left, which is consistent with the literature. This produces the aforementioned effects. However, it is critical to note that the observed quantitative decrease in the total integrated oxidation source term under higher dilution ratios could also be because of the reduction in soot surface availability, rather than only a deficiency in gas-phase OH/O₂ oxidizers. As CO₂ dilution severely suppresses upstream nucleation and HACA surface growth, the total mass and surface area of generated soot drop drastically. Consequently, the smaller magnitude of the integrated oxidation term primarily reflects

this lack of soot particles available for oxidation. Future investigations normalizing these consumption rates per unit of available soot surface area will be valuable to completely isolate the shifting chemical reactivity of the oxidizers from mass-based transport constraints.

It is noteworthy that, although all source terms decline with an increasing dilution ratio, the normalized surface growth term drops more sharply than nucleation as CO₂ dilution rises. The surface growth process is driven mainly by the HACA

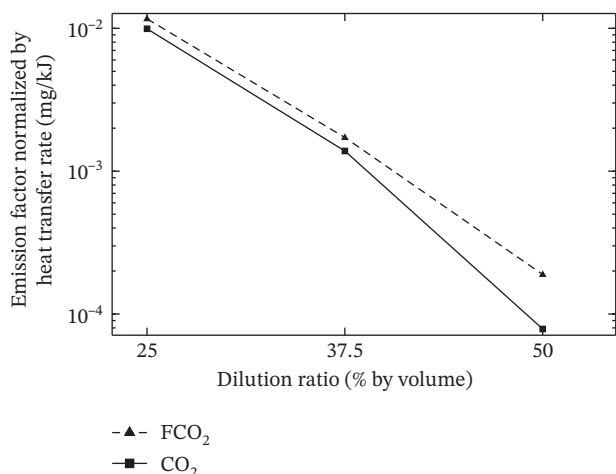


FIGURE 11 | The emission factor normalized by heat transfer rate (mg/kJ).

mechanism, which depends heavily on the concentration of radicals, including H, that decreases as a result of reaction R1. Nucleation, on the other hand, is mostly influenced by pyrene dimerization, whereas the effect of the H radical is less pronounced. Aside from emission reduction performance, the power generated by the combustion process is critical for the direct flame impingement heat transfer application. Because it is anticipated that power decreases as the dilution ratio increases, normalizing the amount of soot emitted with the heat transfer rate (kW) provides a more effective comparison. Since the study focuses on the heat transfer to the surface, the heat transfer rate includes both the heat convection (q_c) and heat radiation (q_r) at the pot surface, which can be expressed by the following governing equation.

$$q = q_c + q_r. \quad (5)$$

Table 4 presents the information on each dilution ratio.

Figure 11 depicts the emission factor normalized by the heat transfer rate on a logarithmic scale variation along the dilution ratio.

Figure 11 shows that when the dilution ratio increases, less soot is produced in both FCO₂- and CO₂-diluted flames. Furthermore, due to the previously indicated chemical impact, CO₂ addition creates fewer emissions at any dilution ratio while providing the same amount of energy. Finally, the chemical effect on suppressing the emission enhances with the dilution ratio increase. The insights gained from this study may benefit the diluted combustion process across various applications, including the design of cookstoves and the optimization of EGR, necessitating a balance between emission reduction and heat transfer rate.

4 | Conclusions

The present work investigates the kinetics of soot emission in a CO₂-diluted turbulent diffusion flame impinging on a cold surface numerically. The study compares the emission behavior with the 0%, 25%, 37.5%, and 50% by volume of CO₂ addition. The results show that at greater dilution ratios, less soot is formed

for the same amount of fuel used as shown by the NSEF. Because dilution affects the chemical behavior of emissions, fictitious CO₂ (FCO₂) is used to calculate the chemical effect. The numerical results suggest that CO₂ chemically influences soot production, with a more pronounced hypothesized effect at higher dilution ratios. It is found that the chemical effect of CO₂ dilution significantly reduces up to 58% more soot than the thermal and dilution effect alone at a 50% dilution ratio. Furthermore, the study indicates that an increase in the CO₂ dilution ratio within a turbulent diffusion impinging flame results in the upstream shift of the peak soot volume fraction towards the nozzle. The sensitivity of soot formation to the dilution ratio is illustrated through an analysis of normalized volume-integrated source terms, revealing that all soot source terms are more effectively suppressed at higher dilution ratios. The source terms of the soot formation processes are then examined. It becomes apparent that all processes are suppressed by the chemical effect of CO₂. This study implements an application-relevant metric by normalizing the emission factor with the heat transfer rate. It offers a valuable assessment metric tailored specifically for optimizing impinging flame dilution applications. Numerous applications could benefit from the findings of this research, including the EGR system and flame-impinging cookstoves. For example, the emission factor normalized by the heat transfer rate might be used to design systems that meet emission regulations while optimizing the heat transfer rate. Future studies should look into how this approach affects the soot PSDF.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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